"BC16" - ANNUAL REPORT OF SALMON STREAM AND SPAWNING GROUNDS

An Historical Review

The Annual Report of Salmon Stream and Spawning Grounds or "BC16", as it was commonly referred to by its form number, had its beginning in the late 1920's. It was a part of a significant reorganization in the Region (then the "Western Division") when, over the period from 1929 to 1931, new personnel were taken on strength, and new reporting and operating procedures were instituted. Except for revisions in 1934, 1948, and 1967, the stream report would change little in over 50 years until the mid-1980's.

LIST OF APPENDICES

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Prior to 1927, spawning reports, like other annuals, took the form of a personal letter to the Supervisor. Runs were described using terms such as "good", "average", "adequate", "satisfactory", "well seeded", etc.. Sockeye producers and a few of the major producers for other species were the target of mention. There was no reporting structure. (see Appendix I - "Report on Spawning Areas - District No. 2, Fall of 1921")

Under instructions from the Chief Supervisor of Fisheries, all three Districts (Prince Rupert, Nanaimo, and New Westminster) were submitting spawning reports on separate forms for each stream "commencing with the season 1929" (Appendix II). District 2 (North of Cape Caution) used this new "Report on Salmon Stream" starting in 1927. (Appendix III)

The "Report on Salmon Stream" became form "BC16" in 1932 (Appendix IV).

In 1934 a new column was added beside "size of run - light, medium, heavy", this being "number on grounds" with accompanying letter ranges:

A	1 - 50	E	500 - 1000	K	10,000 - 20,000
B	50 - 100	F	1000 - 1500	L	20,000 - 50,000
C	100 - 300	G	2000 - 5000	M	50,000 - 100,000
D	300 - 500	H	5000 - 10,000	N	100,000 and over

(Appendix V)

In a letter from Major JA Motherwell, Chief Supervisor of Fisheries, to James Boyd, Supervisor, Prince Rupert, dated January 8th, 1934, the reason for instituting the range system was that the reports "gave no indication of the size of the stream dealt with, or its importance as a spawning area. It is not intended that any effort should be made to tell the actual number but the inspecting officer is so familiar with conditions that he realizes that a small stream well seeded could probably have only one hundred parent salmon, but a larger stream equally well seeded would have perhaps five hundred parent salmon on the spawning grounds." (Appendix VI)

It would appear that this additional information was intended only to give some perspective as to the size of the stream when used in conjunction with the size of the run - light, medium, or heavy. At some time over the years, this letter range began to be construed and used as a more finite estimation of actual escapement numbers. The use of this information for which it was never intended has led to a great deal of criticism of the BC16 in recent years. This has grown to include a general criticism of the methods of spawner enumeration.

Where finite numbers were sought in compiling or using escapement data by stream or by Sub-district (for example, the Stream Catalogue or annual salmon expectations calculation), letter ranges for "total no. on grounds" were split:

E (500 - 1000) became 750

F (1000 - 2000) became 1500

G (2000 - 5000) became 3500

H (5000 - 10,000) became 7500

K (10,000 - 20,000) became 15,000

M (50,000 - 100,000) became 75,000

N (over 100,000) ???

In 1948 the stream report was revised to reflect a few basic physical and biological conditions with the addition of space for comment on:

Physical Conditions of Spawning Grounds

(A) Evidence of erosion and silting

(B) Particulars of scouring of spawning beds

(C) Water levels (low, normal, high, abnormal)

Biological Conditions

(A) Particulars of distribution

(B) Comments re: predators

(C) Evidence of digging up eggs by later spawning fish

The section on obstructions remained and a general comments section was added. (Appedix VII)

In the late 1940's a "Key Stream Program" was undertaken, but was abandoned by the mid-1950's on the basis of costs and benefits. The "Key Stream Reports" (Appendix VIII) were often filed with the BC16. The object of this program was to field monitor selected key streams for overwintering survival of salmon spawn. Winter travel by boat, snow conditions, and water levels posed problems, as did the difficulty of locating redds even in ideal conditions.

In 1958 the BC16 became form "F.381". (Appendix IX)

An important change to note occurred in 1967 with regards to timing. This has implications in gathering and analyzing timing information. Prior to this year, timing information was required in the form of "Dates and duration of run: start, peak, end". This was changed to a column for "Date of arrival in stream", and "Dates of duration of spawning: start, peak, end". (Appendix X)

There was some confusion as to the definition of "run" (pre-67), whether it meant migration through the fishing area, that is, the run timing of that stock, or the run up the river. From personal communication, the definition varied with individual interpretation. The change in 1967 is clear as to arrival in stream, and the start, peak, and end of spawning. What was not clearly defined was what constituted arrival in the stream, either in numbers present, or in light of the fact that many coastal streams might realize the arrival of spawners at the mouth but that their ascent into the stream might be delayed for days or weeks by low water conditions.

In 1984 changes were made to the annual spawning ground report in an attempt to improve the reliability of the final escapement by documenting the individual inspections, survey methods, and stream and weather conditions in greater detail. (Appendix XI)

The following year, a field Stream Inspection Log was introduced to document individual inspections. This included a checklist of stream and weather conditions, methods of survey, percent of spawning area surveyed, and details of numbers live, dead, counted, and estimated. (Appendix XII)

This continues to date of writing (1987).

HR McNairnay
for FEA Wood, Director,
Program Planning and Economics Branch
DFO - Pacific

May 1987

REPORT ON SPAWNING AREAS - DISTRICT No. 2
FALL OF 1921.

QUEEN CHARLOTTE ISLANDS

Yakoun River

There was a small run of Sockeye to this stream which, compared with former seasons, was this year just medium. The run of Cohoe, on the contrary, was heavy, and the run of Red Springs was only medium. The Pinks, of course, only run in the even years to any extent.

Owun Creek

This is a small creek and is frequented by a few Sockeye and some Cohoe. The run was about an average one. The Pinks are expected in large quantities, as usual, during next season.

Sangan Stream

Very few Sockeye enter this stream, but the run of Cohoe was good. The run of Pinks is heavy on alternate years. The spawning grounds are limited.

Naden River

A few creek Sockeye and also some Cohoe run to this stream.

Copper Bay

There was a fair run of Bluebacks at this point and the spawning areas were seeded up to the average. There was also a very good run of Pinks to Copper Bay this year.

Telell River

There was a good run of Cohoe to this stream.

This year it is reported that no damage was done to the spawning beds in the Queen Charlotte Islands area by unusual freshets.

NAAS RIVER DISTRICT

An inspection of this district was made by two officers, one from the Federal and one from the Provincial Department, who have made similar inspections each season for several years past. Apparently there were few Sockeye to be observed on the spawning beds in Meziadon Lake and there was a slight decrease at the Meziaden Falls in comparison with the years 1919 and 1920. The run of Spring Salmon was not as good, but the Cohoe appeared to be more plentiful.

NAAS RIVER DISTRICT (Continued)

There was no inspection made of the spawning areas of the Naas in 1918, but at Meziaden Falls in the years 1908 to 1917 there were always large numbers of Sockeye to be observed. In recent years this satisfactory condition has not existed to the same extent and it would appear, from the information available, that in the Meziaden district the run of Sockeye has diminished very considerably, the season of 1919 bringing no encouragement in this respect.

In the past, inspections of the Naas spawning grounds have been confined to the Meziaden district, but commencing with 1922 it is the intention to extend the inspection to other branches of the Naas, particularly the Bowser Lake district.

Travelling in the Naas district is so hazardous and dangerous as to make it possible for only the most experienced and physically fit men to undertake this work.

This year it is reported that no damage was done to the spawning beds by unusual freshets.

SKEENA RIVER DISTRICT

Fifteen-Mile Creek - Babine Lake

This creek has a splendid spawning area about one-half mile in length and the hatchery fence is constructed at the upper end of the spawning area in order to give the salmon an opportunity of naturally seeding practically all the favourable ground. There was a considerable number of small Sockeye, both male and female, at this point. It is estimated that this creek was much more abundantly seeded this year than last.

Pierre Creek - Babine Lake

There is a splendid spawning area here of about one mile in length and large numbers of spawning Sockeye salmon were seen, besides quite a large school in the deep water at the mouth. The condition of the gravel bottom of the creek showed that a considerable number of Sockeye had already spawned at the time of the inspection, and there were numbers of green fish in the creek which had not yet spawned.

Hatchery Creek - Babine Lake

In this stream there are a number of very fine spawning areas between the mouth and Morrison Lake, its source, a distance of about two and a half miles. The stream is clear of obstructions and large numbers of sockeye salmon were seen between the mouth of the stream and the barricade installed for hatchery purposes at the outlet of Morrison Lake.

Hatchery Creek - Babine Lake (Continued)

The majority of the salmon seen in this stream were large. The first Sockeye were reported on June 18th, but as the hatchery fences were not put in position until July 15th, undoubtedly a considerable number of fish were able to ascend Morrison Lake and Salmon River, at the head of the lake. The seeding in Hatchery Creek is well up to the average.

In connection with the hatchery stream, it is interesting to note that when the hatchery was first established, it was found necessary to go great distances in order to obtain eggs from other creeks, but during recent years, owing to the artificial operations, it is now unnecessary to take eggs from any other point.

Babine River

This river, which empties Babine Lake into the Skeena River, was inspector for a distance of approximately ten miles from its source. Few Sockeye were seen on the many spawning areas observed, but from the appearance of the spawning areas in the lake and its tributaries, it is very evident that large quantities of salmon had ascended the river before the date of inspection. The natural spawning of the river itself, however, would appear to be not entirely satisfactory.

Fulton River - Babine Lake

This stream, for a distance of half a mile from the lake, is practically a slough, but beyond that for about two miles contains splendid spawning areas and here large numbers of Sockeye were observed. In this creek very few Sockeye were seen last season, but this year the supply is satisfactory.

Four-MileCreek - Babine Lake

This is a very small stream. A number of dead Sockeye were observed near its mouth and a few live fish farther up. The local Guardian states, however, that quite a large number of fish have found their way up the stream and that the seeding is better than last year.

Beaver River - Babine Lake

The first three miles from the lake is practically all slough and owing to the water being deep and muddy it is difficult to say whether any salmon were present or not. The principal spawning area is about seven miles up, where Grizzly Creek joins the Beaver. The local Guardian visited this point twice during August, and both times observed large numbers of Sockeye. The previous season very few were seen.

There are also several smaller spawning areas between the numerous log jams in the seven miles and large numbers of Sockeye were also seen at these points.

In every creek entering into Babine Lake quite a number of both Pink and Cohoe salmon were observed, apparently more than usual.

Beaver River - Babine Lake (Continued)

It was noticed that an unusually large number of male Sockeye were marked by gill-nets, but it is difficult to say whether these markings were the result of their having passed through the nets before they got to the Babine River or whether they were due to the operations of the Indians.

To sum up, it would appear that the spawning areas of the Babine district, while not being as well seeded as in the years 1918 and 1919, still show a very satisfactory quantity of spawning Sockeye salmon and well up to the average.

Buckley River

The officer patrolling this district estimates that more Sockeye reached the spawning grounds on the Buckley River this year than was the case last season. At this point there was a big run of Pinks and Cohoe reported, as well as a good run of these two species to the Kispiox River.

Lakelse Lake.

The run of salmon to this point was very disappointing. Approximately 4,306,000 Sockeye eggs were taken for hatchery purposes. Williams Creek, which is the principal Sockeye stream on the lake, was fairly well seeded, and most of the eggs for the hatchery were obtained from this source.

At Fire Warden's and Scullabuchan Creeks a few Sockeye were to be seen, although owing to the runs to these creeks being earlier than others, a number of sockeye got past before the hatchery fences were put in place.

All the spawning areas in this vicinity are free of obstructions.

There was a good run of Pink and Cohoe salmon to the Lakelse Lake district, but generally speaking, the Lakelse Lake district was very poorly seeded. This is no doubt accounted for to a very large extent by the fact that in the winter of 1917-18 a big freshet occurred which washed out the spawning beds and ruined a large percentage of the ova deposited there.

Prince Rupert District

The only Sockeye streams in the vicinity of Prince Rupert which contribute to the Skeena River Sockeye run are the Shawatlans and Cloyah Creeks. The former stream has been closed to Sockeye fishing for a full cycle for the purpose of conserving the run which had become depleted. The results have been very satisfactory and fishing operations to a limited extent will be permitted there this season.

General

There is no doubt that the Indians in the Skeena River watershed, who are taking large quantities of spawning Sockeye salmon off the spawning grounds, are doing much

General (Continued)

damage to the supply of this variety and it is hoped that some arrangement can be made with the Department of Indian Affairs before next spawning season to the end that this practice of the Indians will be discontinued., or at least curtailed. The Sockeye fishing is of far too great a value to permit of its being ruined due to the depredations of Indians and the country would be benefited generally if the Indians were supplied by the Government with other food to take the place of the salmon which they have been accustomed to take off the spawning grounds.

This year it is reported that no damage was done to spawning beds by unusual freshets.

CENTRAL DIVISION

This area extends from the Skeena River south to Calvert Island, but does not include Dean and Burke Channels.

Namu River

The Namu River was fairly well stocked with spawning Sockeye. This point was not extensively fished during the season.

Kisomets Creek in Fisher Channel

A fair number of Sockeye passed up this stream, but the spawning areas are so distant that the officers have not yet been able to include an inspection of same with their other duties.

Port John

A fair number of Sockeye were observed on the spawning beds. Very little sine fishing was done at this point.

Howyet

A fair number of Sockeye passed up the stream during the run, but it was not found possible, owing to lack of facilities, to properly inspect the spawning grounds themselves.

Ship Point on Campbell Island

The number of Sockeye on spawning beds in this vicinity was small. The spawning areas are limited.

McLaughlin Bay

A small number of Sockeye were observed on the spawning beds. Efforts are now being made by means of planting eyed eggs to build up this run and fishing will have to be curtailed or altogether prohibited for a cycle.

Quaquandis

A small number of Sockeye were seen ascending this stream. The spawning grounds themselves were not visited.

Tinkey

Very few Sockeye were observed on the spawning grounds at this point. This stream has been fished quite hard by seines for years and would appear to require some greater protection than it has been receiving.

Tuno

A fair run of Sockeye was observed on the spawning grounds. Very little fishing was done this season by means of seines.

South Point on Millbank Island

In this vicinity a fair number of Sockeye appeared on the spawning beds, which are very limited in extent. On account of the fish being so small, averaging about two pounds, they have not been sought after.

Price Island in Schooner Pass

A very small spawning area, with a fair number of Sockeye.

Mary Cove on Finlayson Channel

Spawning area small, with a fair number of Sockeye.

Cua Cua on Swindle Island

A fair number of Sockeye but apparently requiring more protection.

Tullamass on Princess Royal Island

A fair number of Sockeye on spawning beds but run apparently decreasing.

Aristazable Island Creek

Spawning area small, but small number of Sockeye this season.

North Surf Inlet Creek

A fair number of Sockeye on the spawning beds, but apparently diminishing.

Hartley Bay

Very few Sockeye this season on the spawning beds. This stream runs through an Indian village which may account to some extent for the shortage of spawning fish. The spawning area is limited.

Old Hartley Bay

This stream is evidently showing signs of the intensive fishing for some years past, as few Sockeye were observed on the spawning beds.

Daniels Bay

The run of sockeye observed on the spawning beds was good. No fishing carried on at this point this season.

Kitlum River - Gardner Canal

No fishing was carried on this season and a satisfactory number of Sockeye were observed on spawning beds.

Union Pass on Pitt Island

Fair number of Sockeye on spawning beds.

Deer Point Creek - Banks Island

Fair number of Sockeye found their way to the spawning beds, the heavy rains all summer enabling an unusually large proportion of the run to ascend the stream.

The conditions at Bear Hill Creek, Despair Point, and Endhill Bay, all on Banks Island, and Curtis Inlet Creek and Mikado Bay on Pitt Island, are similar to those reported for Deer Point Creek.

Mink Trap Bay

Some Sockeye were observed at this point, more than in previous years. This season the log jam which had blocked the stream for many years was removed by the Department. This has resulted in the spawning areas being well seeded and should be of much benefit in future years.

Quai Inlet Creek - Princess Royal Island

Spawning area limited and small number of Sockeye observed.

Lowe Inlet Creek on Lowe Inlet

A fair number of Sockeye were observed on the spawning beds.

This year it is reported that no damage was done to the spawning beds in the Central Division by unusual freshets.

BELLA COOLA AND KIMSQUIT

Atnarko River

An inspection this season showed that the work done in clearing the river of obstructions has been productive of excellent results. There was a very satisfactory supply of Spring salmon on the spawning grounds—better than seen on any previous inspection.

The supply of Sockeye on the spawning beds was about fifty per cent of the previous season. A few Pinks and Chums were observed.

Kimsquit River

The supply of Sockeye and Spring salmon on the spawning grounds in this river was found to be very satisfactory.

<u>Burke and Dean Channels</u>

The small streams running into the lower portions of these two channels were found to be fairly well seeded with Chum and Cohoe eggs.

Four years ago, in the Bella Coola and Kimsquit area, the freshets were exceedingly severe and washed out all the spawning beds, doing immense damage. The result of these freshets has been apparent in the cycle year 1921.

<u>RIVERS INLET</u>

While a number of the streams in the Owekano Lake district were well stocked with spawning Sockeye salmon this season, several of the streams were practically failures.

<u>Quap Lake</u>

At this point the run was quite satisfactory.

<u>Genesee Creek</u>

The work done at this point in the way of clearing obstructions resulted in extending the spawning area by approximately fifty per cent. The condition of the beds was very satisfactory, a good supply of spawning fish being observed.

<u>Cheo River</u>

This stream appears to have been a failure this year as practically no Sockeye were observed.

<u>Indian River</u>

The same remarks apply here as in the case of the Cheo River.

<u>Wakwash</u>

The run to this stream was very small compared with that of last season.

<u>Shumahalt River</u>

This is usually one of the best Sockeye spawning streams in the district; but this season the supply of spawning fish was very disappointing.

<u>Nookins</u>

Very few Sockeye were observed here and the season's spawning was practically a failure. This is an early run stream.

<u>Asklum River</u>

The supply of Sockeye salmon on the spawning areas at this point was very good. Owing to the washing out of the hatchery fences, a large number of Sockeye were able to pass up stream and deposit their eggs naturally.

Dallos River

The conditions here were found to be very satisfactory. On the lower reaches large numbers of dead spawned Sockeye were observed on the bottom, and farther up the stream was well seeded.

Markwell River

This stream is not frequented by Sockeye but considerable work was done during the year by the Engineering Department which resulted in diverting the river back into old channels, thereby preventing the Markwell from breaking through to and destroying the spawning beds of Genesee Creek.

Hatchery Creek

There was a fair showing of Sockeye at this point.

On the whole, it would seem that the Rivers Inlet district was moderately well seeded. The early run streams, however, were practically failures.

It is reported that this year no damage was done to the spawning beds by unusual freshets.

SMITHS INLET

The reports from this area are not very encouraging, particularly as the first reports received from the officer at Quashela Creek were to the effect that considerable quantities of Sockeye were passing up stream.

Geluck River

The inspecting officer in previous years has found the spawning bars of this stream covered with Sockeye salmon, but at the date of his inspection this season very few were seen. Although the stream was high and muddy, sufficient evidence appears to be available to warrant the statement that this year's spawning did not compare favourably with the runs of 1918, 1919 and 1920. The showing in the two small streams flowing into the Geluck River was very poor.

Delawah River

The condition of the spawning areas here was found to be more encouraging and apparently the beds were fairly well seeded, although not to the same extent as in previous seasons.

Quay Creek

The run to this point is estimated to be very small, although, owing to the high water, it was difficult to make an accurate estimate.

Docee River

Considerable numbers of Spri g salmon were noticed in this stream. They spawn on the shallows at the foot of the lake. It will be remembered that the drag-seine operations at Quashela Creek were stopped in 1919 and it is expected that in the season 1923 the run to this point will show an improvement.

This year it is reported that no damage was done to spawning beds in the Smiths Inlet district by unusual freshets.

Appendix II - Letter from Chief Inspector of Fisheries initiating use of annual spawning reports for individual streams (dated November 24th, 1928)

LETTER No.

November 24 28.

JAM:P

30-5-1

SPAWNING REPORTS

Dear Mr. Taylor:

I would refer to your letter of the 21st instant, file 41-1, with regard to the above subject. You will observe that a supply of this paper has been forwarded to you. The reports received from your district are, generally speaking, very good when dealing with spawning areas. The overseers appear to show an intelligent interest in the matter. We have a system in Districts Nos. 1 and 2, by which a separate form is used for each stream. As the fishing becomes more intensive and our supervision becomes closer it is imperative that we have a special report on each stream on a separate form. At this office we are arranging a file for each stream in the province which as you will see contemplates a separate form. I would therefore ask you, commencing with the season 1929, to have the officers forward the special forms in connection with each stream, and also at the end of the season cover spawning matters generally in his district with a narrative report, not of course dealing particularly with each stream, but in a general way. A supply of forms is being forwarded to you.

Yours faithfully

J.A. Motherwell
Chief Inspector of Fisheries

E.G. Taylor, Esq.,
Inspector of Fisheies,
Nanaimo, B.C.
Copy handed G.F. Found

REPORT ON SALMON STREAM

District No., 2........ Fishery Officer. C. T. Bainbridge

Name of Stream.....Slatse Creek..... Date of Report. 30th; Sept; 1927.

LOCATION:

Flowing into.........South Roscoe Inlet.........miles, ... (N.S.E.or W.)

.... (known location)

NOTE: If possible make sketch on back of this form showing location of stream with reference to some known point or bay, and also showing location of obstructions in stream.

PARTICULARS WITH REGARD TO RUNS OF SALMON TO STREAM COVERED BY THIS REPORT

Species	Date of Arrival	Size of run - Heavy, Medium or Light.	Remarks
~~Sockeye~~			
~~Cohoe~~			
~~Springs~~			
Pinks	16th; Aug.	Light	Poorly seeded
Chums	25th, "	Still running	
~~Steelhead~~			
~~Bluebacks~~			

NOTE: Draw lines through names of fish that do not frequest stream.

OBSTRUCTIONS IMPASSABLE TO ASCENT OF FISH (IF NO IMPASSABLE OBSTRUCTIONS, STATE SO.):-

(1) Nature of obstruction...

(2) Distance from mouth of stream...................................

(3) Do you recommend that the obstruction be removed? If so, state your reasons, and describe nature and extent of spawning grounds above obstruction:

....

...................- NIL -...

Appendix IV - Report on Salmon Stream becomes official form "BC16" in 1932

B. C. 16. REPORT ON SALMON STREAM

District No. 2 Fishery Officer G. S. Reade
Name of stream KLATSE Date of report Nov. 22/32

= =

LOCATION:
Flowing into ROSCOE INLET — miles at Indian
(N.S.E. or W.)
Reserve No. 5
(known location)

NOTE: If possible make sketch on back of this form showing location of stream with reference to some known point, and also showing location of obstructions in stream.

= =

PARTICULARS WITH REGARD TO RUNS OF SALMON TO STREAM COVERED BY THIS REPORT

Species	Date of arrival	Size of run - heavy, medium or light.	Compare with brood year	Remarks
~~Sockeye~~				
Cohoe		(unobserved	probably	light)
~~Springs~~				
Pinks	Aug 28	medium	Heavy	
Chums	Aug 21	V. Heavy	V. Heavy	
Steelhead	March	few	unobserved	

NOTE: Draw lines through names of fish that do not frequent stream.

= =

OBSTRUCTIONS IMPASSABLE TO ASCENT OF FISH:
(1) Nature of obstruction nil
(2) Distance from mouth of stream
(3) Do you recommend that the obstruction be removed? If so, state your reasons, and describe nature and extent of the spawning grounds above obstruction:

OBSTRUCTIONS FORMING BUT NOT IMPASSABLE TO ASCENT OF FISH:
(1) Nature of obstruction
(2) Distance from mouth of stream
(3) Do you recommend that the obstruction be removed? If so, state your reasons, and describe nature and extent of the spawning grounds above obstruction:

G. S. Reade
Fishery Inspector

= =

Appendix V - Modification to the BC16 in 1934 to include letter ranges

B .16

REPORT ON SALMON STREAMS

District No..2..Fishery Officer..G.S.Reade.........Date..Sept.13/37.

Name of stream...KKlatse..R.....Flowing into.Roscoe.Inlet...........

miles.............Known location....at.Indian.Reserve.No..5.........
(N.S.E. or W.)

Note: If possible make sketch on back of this form showing the location of stream, with reference to some known point, and also showing location of obstructions in stream.

= =

PARTICULARS WITH REGARD TO RUNS OF SALMON TO STREAM COVERED BY THIS REPORT

Species	Date of Arr.	No. on grounds (See note below)	Size of run-Heavy, Medium, or Light	Compare with brood year	Remarks
~~Sockeye~~					
~~Cohoe~~					
~~Springs~~					
Pinks	Aug.	H	v. heavy	H	
Chums	Sept.	L	heavy	medium	
~~Steelhead~~					

Note: Draw lines through names of fish that do not frequent stream.

= =

OBSTRUCTIONS IMPASSABLE TO ASCENT OF FISH.

(1) Nature of obstruction ....fall..........

(2) Distance from mouth of stream ....2 miles..........

(3) Do you recommend that the obstruction be removed? .....no.....
If so, attach report stating your reasons, and describe nature and extent of spawning grounds above obstruction.

= =

OBSTRUCTIONS FORMING BUT NOT IMPASSABLE TO ASCENT OF FISH.

(1) Nature of obstruction ....nil..........

(2) Distance from mouth of stream ....

(3) Do you recommend that the obstruction be removed? ....
If so, attach report stating your reasons, and describe nature and extent of the spawning grounds above obstruction.

= =

NOTE: Estimate number of parent fish on spawning grounds and indicate by placing letter in column provided to show approximate number, thus:

1-50 A; 50-100 B; 100-300 C; 300-500 D;

500-1000 E; 1000-2000 F; 2000-5000 G;

5000-10000 H; 10000-20000 K; 20000-50000 L;

50000-100000 M; Over 100000 N.

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Appendix VI - Letter from Chief Inspector of Fisheries explaining use of new letter ranges and intent (dated January 8th, 1934)

January 8, 34.

30-8-1
AM/D

Dear Sir,

Form B. C. 16, Report on Salmon Streams, which has been in use up to date has been found of particular value in dealing with the conditions on the various streams but the information which was supplied gave no indication of the size of the stream dealt with or its importance as a spawning area. With a view to remedying this defect and making these forms of still greater value, a revised form has been prepared in which it is intended that the approximate number of parent salmon on the spawning grounds should be indicated. It is not intended that any effort should be made to tell the actual number but the inspecting officer is so familiar with conditions that he realizes that a small stream well seeded could probably have only one hundred parent salmon but a larger stream equally well seeded would have perhaps five hundred parent salmon on the spawning grounds.

A limited supply of this form has been forwarded to you, the intention being that these should be submitted to the various inspecting officers with, of course, your instructions in connection with completing the forms.

Kindly advise in the near future the number of these forms that you will require for use during the season of 1934.

Yours truly,

For J. A. Motherwell
Chief Supervisor of Fisheries.

J. Boyd, Esq.,
Supervisor of Fisheries,
Prince Rupert, B. C.

Appendix VII - Modification to the BC16 in 1948 to include new sections covering physical and biological conditions, and general comments.

B.C. 16

K52

SALMON STREAM SPAWNING REPORT

DISTRICT NO. 2, B.C. FISHERY OFFICER R.C.Edwards YEAR 1948.

NAME OF STREAM Klatse River (MAP NAME) ______ (LOCAL NAME) FLOWING INTO Roscoe Inlet

DATES ON WHICH STREAM INSPECTED Sept.7th, to Oct. 27th. 1948.

NOTE:- A SKETCH OF THIS STREAM IS REQUIRED ON THE BACK OF THIS FORM, SHOWING IN ADDITION TO RELEVANT DATA SUCH AS LOCATION OF OBSTRUCTIONS, GENERAL OUTLINE OF TOPOGRAPHY ALONG THE STREAM, PORTIONS OF STREAM BED WHERE SPAWNING OCCURS, ETC., ITS LOCATION IN RELATION TO SOME KNOWN POINT. WHEN SUCH SKETCH HAS ONCE BEEN MADE AVAILABLE, IT MAY BE REFERRED TO IN FOLLOWING REPORTS.

PARTICULARS OF SPAWNING AND SPAWNING CONDITIONS

		SOCKEYE	SPRINGS	COHOE	PINKS	CHUMS	STEELHEADS
1. DATES OF DURATION OF RUN	START	xx	xx	xx	Aug.20	Sept.1	xx
	PEAK	xx	xx		Sept.25	Sept.25	xx
	END	xx	xx		Oct.5	Oct.30	xx
2. TOTAL NUMBER ON GROUNDS					K.	L.	
3. SIZE OF RUN-HVY.MED.ET.					Med.	Hvy.	
4. COMPARE WITH TOTAL NUMBER FOR BROOD YEAR USING SYMBOL					L.	1944 K. 1945 H.	
5. GIVE SEX RATIO IN PERCENTAGES	MALE				Not observed.		
	FEMALE						
	JACKS						

NOTE: DRAW LINES THROUGH NAMES OF SALMON THAT DO NOT FREQUENT THIS STREAM.

6. PHYSICAL CONDITION OF SPAWNING GROUNDS

(A) EVIDENCE OF EROSION AND SILTING - GIVE EXTENT OR % STREAM BED AFFECTED Condition satisfactory.

(B) PARTICULARS OF SCOURING OF SPAWNING BEDS OR CHANGE IN COURSE OF STREAM Nil

(C) WATER LEVELS (LOW, NORMAL, HIGH, ABNORMAL). IF ABNORMAL, DETAILS SHOULD BE GIVEN Water levels generally normal during spawning.

7. BIOLOGICAL CONDITIONS

(A) PARTICULARS OF DISTRIBUTION OF SPAWNING SALMON OVER THE STREAMBED Well concentrated throughout length of stream available.

(B) COMMENTS RE PREDATORS Usual seagulls, wildfowl and eagles.

(C) EVIDENCE OF DIGGING UP OF EGGS BY LATER SPAWNING FISH Nil.

8. OBSTRUCTIONS

(A) PASSABLE OR IMPASSABLE Impassable

(B) NATURE OF OBSTRUCTION Falls

(C) DISTANCE FROM MOUTH OF STREAM about 1200 yards.

(D) DO YOU RECOMMEND THAT THE OBSTRUCTION BE REMOVED? No.
(IF SO, ATTACH REPORT STATING YOUR REASONS AND DESCRIBE NATURE AND EXTENT OF THE SPAWNING GROUNDS ABOVE OBSTRUCTION.)

9. COMMENTS ON ANY OTHER CONDITIONS AFFECTING THIS STREAM This river carries large Pink and Chum hatchery.

10. NOTE: ESTIMATE NUMBER OF PARENT FISH ON SPAWNING GROUNDS AND INDICATE BY PLACING LETTER IN COLUMN PROVIDED TO SHOW APPROXIMATE NUMBER, THUS:

1-50	A	300-500	D	2000-5000	G	20,000-50,000	L
50-100	B	500-1000	E	5000-10000	H	50,000-100,000	M
100-300	C	1000-2000	F	10,000-20,000	K	OVER 100,000	N

(ADDITIONAL PAPER MAY BE USED WHERE NECESSARY)

8 m

P.B.S. 1.

Key Stream Report

Area No. # 7 Stream Klatse River

Date Dec. 8th 19 52 Init. H. Burrow

Temperature. Air 36.5 °F. Water 41 °F.

Time of Reading: A.M. 10:50 P.M.

Gauge Reading 4 in. in. Stream Dry: No X Yes

Depth of Snow none present inches

Depth of Frost in Gravel none present inches

Flood Damage since last Survey: No X Yes

Light Moderate Severe

Explanation:

No. Redd Samples 1 Species Pinks

	Eggs	Alevins
Number Live	2	
Number Dead	23	

Salmon Observed: Check if present

	Sock.	Spr.	Coho	Pink	Chum	Trout
Fry:						
Yearlings:						

Fry or Alevins Stranded: No Yes

Remarks: All dead eggs not eyed
One of live eggs broke
and alevin headed back
down again into gravel.

Appendix IX - BC16 becomes form F381 in 1958

F. 381

SALMON STREAM SPAWNING REPORT - PACIFIC AREA

DISTRICT NO. 2 FISHERY OFFICER R.R. Mallory YEAR 1958

NAME OF STREAM Klatse Creek FLOWING INTO Roscoe Inlet
Map Name Local Name

DATES ON WHICH STREAM INSPECTED August 20th - Sept. 27th - October 2nd

NOTE:
A sketch of this stream is required on the back of this form, showing in addition to relevant data such as location of obstructions, general outline of topography along the stream, portions of stream bed where spawning occurs, etc., its location in relation to some known point. When such sketch has once been made available, it may be referred to in following reports.

PARTICULARS OF SPAWNING AND SPAWNING CONDITIONS

	~~Sockeye~~	~~Springs~~	Cohoe	Pinks	~~Steelhead~~	Chums
1. Dates of duration of run) Start				August 20		Sept. 15
) Peak	Odd cohoe seen.			Sept.		Oct. 1
) End				Oct. 1		Oct. 10
2. Total number of grounds				"G"		"K"
3. Size of run - hvy. med. lt.				Light		Medium
4. Compare with total number for brood year using symbol				"L"		"K"
5. Give sex ratio in percentages) Male				50%		50%
) Female				50%		50%
Jacks						

NOTE: Draw lines through names of salmon that do not frequent this stream.

6. PHYSICAL CONDITION OF SPAWNING GROUNDS
   (A) Evidence of Erosion and Silting - Give Extent or % Stream Bed Affected
   None
   (B) Particulars of Scouring of Spawning Beds or Change in Course of Stream
   None
   (C) Water Levels (Low, Normal, High, Abnormal). If Abnormal, details should be given
   High at each inspection trip

7. BIOLOGICAL CONDITIONS
   (A) Particulars of Distribution of Spawning Salmon over the Streambed Pinks all spawned in tidal influence and lower grave. Chums over entire river.
   (B) Comments re Predators
   Bear damage light.
   (C) Evidence of Digging up of Eggs by Later Spawning Fish
   None

8. OBSTRUCTIONS
   (A) Passable or Impassable Impassable
   (B) Nature of Obstruction Falls
   (C) Distance from Mouth of Stream 1 mile
   (D) Do you Recommend that the Obstruction be Removed? No.
   (If so, attach report stating your reasons and describe nature and extent of the spawning grounds above obstruction.)

9. COMMENTS ON ANY OTHER CONDITIONS AFFECTING THIS STREAM
   Adequate water levels throughout spawning period. In fact large percentage of chums spawned at very high water which may cause some loss of fry in spring 1959.

10. NOTE: Estimate Number of Parent Fish on Spawning Grounds and Indicate by Placing Letter in Column Provided to Show Approximate Number: Thus

1 - 50 A	300 - 500 D	2000 - 5000 G	20000 - 50000 L
50 - 100 B	500 - 1000 E	5000 - 10000 H	50000 - 100000 M
100 - 300 C	1000 - 2000 F	10000 - 20000 K	* Over 100000 N

* Where letter "N" used it is requested approximate number of parent fish on spawning grounds be shown.

Appendix X - Modification to F381 in 1967-68 reflecting changes in the reporting of the timing of arrival and spawning

DEPARTMENT OF FISHERIES
PACIFIC REGION

CANADA

DISTRICT NO. C.D. 7

YEAR 1968

ANNUAL REPORT OF SALMON STREAM AND SPAWNING GROUNDS

NAME OF STREAM (MAP NAME)	(LOCAL NAME)
Klatse River	

FLOWING INTO	DATES STREAM INSPECTED
Roscoe Inlet	Aug. 4, 11, 18, 26, Sept. 7, 20, Oct. 11.

NOTE: A sketch of this stream is required on the back of this form, showing in addition to relevant data such as location of obstructions, general outline of topography along the stream portions of stream bed where spawning occurs, etc., its location in relation to some known point. These sketches should be brought up to date every five years.

PARTICULARS OF SPAWNING & SPAWNING CONDITIONS — (Draw lines through names of salmon that do not frequent this stream.)

SPECIES	ARRIVAL IN STREAM	DATES OF DURATION OF SPAWNING			TOTAL NO. ON GROUNDS	SIZE OF RUN			BROOD YEAR SYMBOL	GIVE SEX RATIO IN %		
		START	PEAK	END		HVY.	MED.	LT.		M	F	JACKS
SOCKEYE												
SPRINGS												
COHOE	Sept. 1	Sept. 30	Oct 15	Oct. 30	B (50)			X	N.O.	50	50	N.O.
PINKS	Aug 25	Sept. 1	Sept 20	Oct. 20	M (50M)	X			M	50	50	
STEELHEAD												
CHUMS	Sept. 1	Sept 5	Sept 20	Oct 25	L (25M)	X			H	50	50	

NOTE: Estimate Number of Parent Fish on Spawning Grounds and indicate by placing letter in Column provided to show approximate number: Thus

1 - 50 A	300 - 500 D	2000 - 5000 G	20000 - 50000 L
50 - 100 B	500 - 1000 E	5000 - 10000 H	50000 - 100000 M
100 - 300 C	1000 - 2000 F	10000 - 20000 K	*Over 100000 N

*Where letter "N" used it is requested approximate number of fish on spawning grounds be shown.

PHYSICAL CONDITION OF SPAWNING GROUNDS

(A) Evidence of Erosion and Silting - Give Extent or % Stream Bed Affected None observed.

(B) Particulars of Scouring of Spawning Beds or Change in Course of Stream Slight in middle reaches.

(C) Water Levels (Low, Normal, High, Abnormal). If Abnormal, details should be given Low in August, high most of September, very high with frequent flooding during October.

BIOLOGICAL CONDITIONS

(A) Particulars of Distribution of Spawning Salmon over the Stream Bed Pinks and chums well distributed to falls and coho lightly scattered throughout.

(B) Comments re Predators Moderate in lower reaches by wolves.

(C) Evidence of Digging up of Eggs by Later Spawning Fish Light indications throughout.

OBSTRUCTIONS

(A) Passable or Impassable Impassable.

If Nil, indicate distance from mouth to furthest point of access

(B) Nature of Obstruction Falls.

(C) Distance from Mouth of Stream One mile.

(D) Do you recommend that the Obstruction be removed?
(If so, attach report stating your reasons and describe nature and extent of the spawning grounds above obstruction)

COMMENTS ON ANY OTHER CONDITIONS AFFECTING THIS STREAM

John A. Macdonald
FISHERY OFFICER

F-381

DEPARTMENT OF FISHERIES AND OCEANS

ANNUAL REPORT OF SALMON STREAMS AND SPAWNING POPULATIONS

STREAM IDENTIFICATION

Watershed code	91-4200
Gazetted name (mapname)	CLATSE CREEK
First local name	KLATSE RIVER
Second local name	
Flows into	Roscoe Inlet

Note: Please correct any stream identification data that is wrong.

District No.	Subdistrict No.
7	07
Statistical Area	Management Area
07	

	Month	Day	Year
Date first inspected	07	20	84
Date last inspected	09	13	84
Total no. of inspections	15		

SPAWNING RUN TIMING AND ESTIMATED NUMBER (instructions on flip side)

1 SPECIES		2 ARRIVAL IN STREAM (Month Day)	3 DATES OF DURATION OF SPAWNING START (Month Day)	PEAK (Month Day)	END (Month Day)	4 NO. OF OBSER.	5 METHODS	6 RELIABILITY	7 EST. TOT. NO. ON GROUNDS
SOCKEYE	1								
	2								
COHO	1								
	2								
PINK	1	Aug 10	E. Sept.	M. Sept	L. Sept	9	A, F	A	12000
	2								
CHUM	1	Aug 10	E. Sept.	M. Sept.	L. Sept	6	A, F	A	3000
	2								
CHINOOK	1								
	2								
STEELHEAD	1								
	2								

Appendix XI - Revised "Annual Report of Salmon Stream and Spawning Grounds" in 1984 to "Annual Report of Salmon Streams and Spawning Populations (G. Serbic, Pacific Biological Station)

CONDITIONS

Mark box for unusual conditions.

- ☐ (A) Enhancement or intense biological activities.
- ☐ (B) Unusual mortalities.
- ☐ (C) Obstructions or changes in habitat with recommendations.
- ☐ (D) Large variations in sex ratio or unusual number of jacks.
- ☐ (E) Unusually high or low water flow level during spawning period.

ADDITIONAL COMMENTS

PHYSICAL CONDITION OF SPAWNING GROUNDS

(A) Evidence of erosion and silting. Give extent or percent of stream bed affected. —

(B) Particulars of scouring of spawning beds or change in course of stream: —

(C) Water levels (low, normal, high, abnormal). If abnormal, details should be given: Water level high and no count possible Sept. 12

FFO 841

BIOLOGICAL CONDITIONS

(D) Particulars of distribution of spawning salmon over the stream bed. —

(E) Comments on predators. —

(F) Evidence of digging up eggs by later spawning fish. —

OBSTRUCTIONS

(G) Passable or impassable. Imp

If nil, indicate from mouth to furthest point of access.

(H) Nature of obstruction. Falls

(I) Distance from mouth of stream. 1.5 Km

(J) Do you recommend that the obstruction be removed? No

(If so, attach report stating your reasons and describe nature and extent of the spawning grounds above obstruction.

COMMENTS ON ANY OTHER CONDITIONS AFFECTING THIS STREAM

(K)

Return this form to:
Salmon Escapement System Co-ordinator
Pacific Biological Station
Nanaimo, B.C. V9R 5K6

STEFAN BECKMANN
Fishery Officer/Person preparing report

Stefan Beckmann
Signature

INSTRUCTIONS For "Spawning Run Timing & Estimated Number"

1 Provision is made for two spawning runs per species. If only one run exists, use Line 1.

2 Date entry: a) Month: enter first three letters (Aug) or (Oct)
b) Day: enter date (12) or (04)
3 or enter letter codes as follows - (A) 1-10th (B) 11-20th (C) 21-31st

4 Number of times each species is present in stream during inspection.

5 Inspection method used. Enter up to 4 methods per species.

A - fixed wing A/C
B - helicopter
C - stream bank
D - boat
E - fence
F - stream walk
G - other (enter details in section (K))

6 Reliability of spawning population estimate (based on conditions and number of stream visits).
A - high B - average C - low

7 a) Enter best estimate of total annual escapement.
b) If species expected but none observed, enter: NO
c) If species present but number unknown, enter: UNK

F
B

DEPARTMENT OF FISHERIES AND OCEANS

ANNUAL REPORT OF SALMON STREAMS AND SPAWNING POPULATIONS

STREAM IDENTIFICATION

Watershed code	
Gazetted name (mapname)	Clatse Creek
First local name	
Second local name	
Flows into	Roscoe Inlet

Year: 1985	
District No. 7	Subdistrict No. 7
Statistical Area 7-16	Subdistrict Name Bella Bella

DATES OF INSPECTION

Month	Day	Month	Day
July	19	Sept	5
Aug.	17	Sept.	6
Aug.	21	Sept	11
Aug.	29	Sept.	17
Aug.	29 (Flight)	Sept	30
Sept. 4			

SPAWNING RUN TIMING AND ESTIMATED NUMBER (instructions on flip side)

1 SPECIES		2 ARRIVAL IN STREAM (Month Day)	3 DATES OF DURATION OF SPAWNING START (Month Day)	PEAK (Month Day)	END (Month Day)	4 NO. OF OBSER.	5 METHODS	6 RELIABILITY	7 EST. TOT. NO. ON GROUNDS	8 OPTIMUM ESCAPEMENT
SOCKEYE	1	E. Sept.	M. Sept			2	1	2	~~4~~ 11	0
	2									
COHO	1	L. Sept.				1	1	2	8	150
	2									
PINK	1	Aug. 29	M. Sept	L. Sept		8	1, 3	4	13,600	30000
	2									
CHUM	1	Sept. 4	M. Sept.	L. Sept.		7	1,3	4	2,540	10000
	2									
CHINOOK	1									
	2									

UNUSUAL CONDITIONS

MARK BOX FOR UNUSUAL CONDITIONS
- ☐ (A) Enhancement or intense biological activities.
- ☐ (b) Unusual mortalities.
- ☐ (C) Obstructions or changes in habitat with recommendations.
- ☐ (D) Large variations in sex ratio or unusual number of jacks.
- ☐ (E) Unusually high or low water flow level during spawning period.

ADDITIONAL COMMENTS

PHYSICAL CONDITION OF SPAWNING GROUNDS

(A) Evidence of erosion and silting. Give extent or percent of stream bed affected ....

(B) Particulars of scouring of spawning beds or change in course of stream ....

(C) Water levels flow, normal, high, abnormal. If abnormal, details should be given Sept. 4. Water level improved, but still low – just passable

BIOLOGICAL CONDITIONS

(D) Particulars of distribution of spawning salmon over the stream bed ....

(E) Comments on predators ....

(F) Evidence of digging up eggs by later spawning fish ....

(G) New obstructions (nature and recommendations) ....

COMMENTS ON ANY OTHER CONDITIONS AFFECTING THIS STREAM

(K) ....

Stefan Beckmann
Signature

STEFAN BECKMANN
Fisher Officer / Person Preparing Report

FP0699/BC16's

INSTRUCTIONS FOR SPAWNING RUN TIMING AND ESTIMATED NUMBER

1 Provision is made for two spawning runs per species. If only one run exists, use Line 1.

2 Date entry: a) Month: enter first three letters (Aug) or (Oct)

3 b) Day: enter date (12) or (04)
or enter letter codes as follows — (A) 1 - 10th (B) 11 - 20th (C) 21 - 31st

4 Number of times each species is present in stream during inspection.

5 Inspection method used. Enter up to 4 methods per species.

1 Walk	4 Helicopter	7 Strip Counts	
2 Float	5 Redd counts	8 Dead Pitch	10 Other
3 Plane	6 Spot Check	9 Tag Recovery	

6 Reliability of spawning population estimate (based on conditions and number of stream visits).

Low 1 2 3 4 5 High

7
a) If the stream has been inspected, enter best estimate of total annual escapement.
b) If the stream has been inspected, but no fish were seen even though water conditions would permit enumeration, enter N.O. (None Observed).
c) If juvenile fish only were observed, enter J.P. (Juveniles Present)
d) If the enumerator(s) observed indications that an inspected stream was frequented by fish, but were unable to make an estimate because of water conditions, enter U.K. (Unknown).
e) If the stream was not inspected, for whatever reason, enter N.I. (Not Inspected)

8 Enter if available

(BACK OF PRECEDING PAGE)

Appendix XII - Stream Inspection Log introduced in 1985

STREAM INSPECTION LOG

STREAM NAME ______________________ OBSERVER ______________________

DATE (Year / Month / Day) [ ] [ ] [ ] TIME (Start / Stop) [ ] [ ]

NOTE - * Means comment on reverse side

SECTION INSPECTED ______________________

METHOD	WATER CONDITIONS	WATER LEVEL	WATER CLARITY	WIND MPH	SKY	FISH COUNTABILITY
1 Walk 2 Float 3 Plane 4 Helicopter 5 Redd counts 6 Spot Check 7 Strip Counts 8 Dead Pitch 9 Tag Recovery 0 Other *	1 Clear 2 Tea 3 Glacial Silt 4 Muddy 5 Slightly Turbid 6 Iced 7 Other *	1 Extremely low 2 Below normal 3 Normal 4 Above Normal 5 Flood	1 6" 2 6"-2' 3 2'-5' 4 5'	1 0-5 2 5-10 3 10-20 4 20-30 5 30	1 Sunny 2 Partly cloudy 3 Cloudy 4 Overcast 5 Rain 6 Heavy	Nil Poor Fair Good Excellent

	SOCKEYE	COHO	PINK	CHUM	CHINOOK
NUMBER COUNTED NO. LIVE					
NO. DEAD					
TOTAL					
% NEW					
% PAIRED					
(Y/N) JUV. PRES.					
Estimated total fish in stream ADULTS					
TTD					
JACKS					
RELIABILTY LOW 1 2 3 4 5 HIGH					
ACTIVE SPAWNING (Start Peak end; MM DD)					

Distribution of fish ______________________

PRE-SPAWNING DEATHS:

Number of fish: [ ]

Cause of mortality: ______________________

SILTING * YES [ ] NO [ ]	EROSION * YES [ ] NO [ ]	OBSTRUCTIONS * YES [ ] NO [ ]

REV. DEC 85

OBSTRUCTIONS (expecially previously unreported or changing ones:)

Type: ______________________________

____

Locations: ______________________________

____

Severity: ______________________________

____

Recommended action: ______________________________

____

SILTING, EROSION: ______________________________

____

COMMENTS AND MAP (To update information): (e.g., fry abundance, predators, habitat information, size of fish, air temperature, water temperature, etc.)

(BACK OF PRECEDING PAGE)